Impact of Social Media Usage on Mental Health

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Introduction

A dataset containing behavioral and psychological data from 10,000+ social media users across multiple platforms. Each record includes user demographics (age, gender, location), social media usage patterns (daily time spent, platforms used, posting frequency, content consumed), and mental health indicators (stress levels, anxiety scores, depression index, sleep quality, self-esteem ratings, etc.). Additional variables may include engagement metrics (likes, comments, shares), online interactions, and digital well-being practices.

Problem Statement

The "Threshold" Analysis: Find the exact point where more time = more stress. Use a Scatter Plot with a Trend Line. You plot "Daily Hours" on the X-axis and "Self-Reported Anxiety Score" on the Y-axis. Identify if there is a "U-shaped" curve where moderate use is healthy, but extremely low or high use is negative.

Time-of-Day Impact: Does when you use it matter more than how long? Create a Heatmap showing average stress levels across different hours of the day. Data likely shows that 2 hours of use in the afternoon has a different mental health impact than 2 hours of use after midnight.

Platform Comparison: Are all apps created equal? Use A/B testing logic or T-Tests. Compare the mean well-being scores of users who primarily use "visual-heavy" apps (Instagram/TikTok) versus "text-heavy" apps (Reddit/X).

Business Question

- Which social media platforms are associated with the highest and lowest average mental health scores
- Compare active vs passive usage (posting vs scrolling).
- Which behavior is more strongly associated with negative mental health outcomes?
- Analyze engagement metrics: do higher numbers of likes/comments correlate with better self-esteem, or increased anxiety?
- Examine sleep patterns: how does late-night social media usage affect sleep quality and mental health scores?